Joseph Kreitz

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EDUCATION

Massachusetts Institute of Technology, *Cambridge, MA* 2018-May 2025
- PhD in Biological Engineering (Advisor: Feng Zhang)
- Topic: Engineering bacterial protein injection systems for therapeutic delivery

Duke University, *Durham, NC* 2014-2018
- B.S., Biology, High Distinction (Advisor: Lingchong You)
- Topic: Rational design of phage cocktails to enhance antimicrobial efficacy

RESEARCH EXPERIENCE

Life Sciences Research Foundation (LSRF) Fellow (Advisors: Min Dong and George Church) 2026-Present
Boston Children's Hospital and Harvard Medical School
- Identifying and engineering extracellular contractile injection systems (eCISs)
- Exploring related CISs for delivery (myophages, pyocins, T6SS)

Postdoctoral Fellow (Advisors: Min Dong and George Church) 2025-2026
Boston Children's Hospital and Harvard Medical School

Graduate Research Assistant (Advisor: Feng Zhang) 2019-2025
Broad Institute of MIT and Harvard
- Demonstrated the first eCIS activity in human cells and mice
- Elucidated the mechanisms of eCIS targeting and cargo loading
- Invented a modular *in vitro* functionalization strategy for eCISs called SPEAR

Graduate Rotation Student (Advisor: Timothy Lu) 2018-2019
Department of Biological Engineering, MIT
- Developed methods for retargeting phage T3 against novel bacterial hosts

NSF REU Fellow (Advisor: Ryland Young) 2016
Center for Phage Technology, Texas A&M University
- Isolated 5 phages for a national project to treat a patient with bacteremia

Undergraduate Research Assistant (Advisor: Lingchong You) 2015-2018
Department of Biomedical Engineering, Duke University
- Developed a method for optimizing phage cocktails based on phage synergy

PUBLICATIONS & PATENTS

1. **Kreitz, J.**, Yang, V., Friedrich, M.J., Pham, J., Macrae, R.K., Zhang, F. (2026). Targeted delivery of diverse biomolecules with engineered bacterial nanosyringes. *Nature Biotechnology* 44, 1121–1125.
2. **Kreitz, J.**, Friedrich, M.J., Guru, A., Lash, B., Saito, M., Macrae, R.K., Zhang, F. (2023). Programmable protein delivery with a bacterial contractile injection system. *Nature* 616, 357–364.
   - *Nature* News & Views, *Nature* Podcast, *Scientific American*, Phys.org, *Freethink*, *GEN*
   - Top 4% of all *Nature* papers of similar age by engagement (top 0.09% of all papers)
3. Zhang, F., **Kreitz, J**. Cell-type-specific targeting contractile injection system. Patent no. WO/2023/158486 (2023).

AWARDS & HONORS

2026 Life Sciences Research Foundation (LSRF) Fellowship, *Boston Children's Hospital/Harvard Medical School*
2026 Ruth L. Kirschstein NRSA Individual Postdoctoral Fellowship (F32), Score: 14 (top 1%), *Declined for LSRF*
2020 Yang-Tan Graduate Fellowship, *Massachusetts Institute of Technology*
2018 Presidential Graduate Fellowship, *Massachusetts Institute of Technology*
2018 Graduation with High Distinction, *Duke University* (top 10% of undergraduate theses)
2017 Dean's Summer Research Fellowship, *Duke University*
2016 NSF REU Fellowship in Biochemistry, *Texas A&M University*
2015 Howard Hughes Research Fellowship, *HHMI/Duke University*
2013 General Member, *Mensa*

SELECTED PRESENTATIONS

2026 "Targeted therapeutic delivery with bacterial contractile injection systems." *Lakeside Protein Toxins and Effectors 2026*, Southbridge, MA. (Talk) (Best talk award - postdoctoral level)
2025 "Targeted delivery of diverse biomolecules with engineered bacterial nanosyringes." *GCM Invited Speaker Series*, UMass Chan Medical School, Worcester, MA. (Invited Talk)
2023 "Programmable protein delivery with a bacterial contractile injection system." *Genome Engineering Symposium*, Harvard Medical School, Boston, MA. (Invited Talk)
2023 "Programmable protein delivery with a bacterial contractile injection system." *Broad Institute Retreat*, Boston, MA. (Invited Talk)
2023 "Programmable therapeutic delivery with a bacterial nano-syringe." *KCA Novel Technologies Symposium*, Sydney, Australia. (Invited Keynote Talk)
2023 "Programmable protein delivery with a bacterial injection system." *Mammalian Synthetic Biology Workshop (mSBW)*, San Jose, CA. (Talk)
2022 "Protein delivery with a bacterial contractile injection system." *Yang-Tan Centers Retreat*, Cambridge, MA. (Talk)
2022 "Protein delivery with a bacterial contractile injection system." *MIT Biochemistry and Toxicology Seminar*, Cambridge, MA. (Talk)
2018 "Interaction network optimization improves the antimicrobial efficacy of phage cocktails." *ACC Meeting of the Minds*, Boston, MA. (Poster)
2017 "Interaction network optimization improves the antimicrobial efficacy of phage cocktails." *NC ASM Annual Meeting*, Raleigh, NC. (Poster) (Best Poster Award - top 5 posters of 70)
2016 "Hunting for phage against pandrug-resistant *Acinetobacter baumannii* clinical isolates." *Biochemistry REU Research Symposium*, College Station, TX. (Talk)
2015 "Characterization of bacterial conjugation *in vivo* using *C. elegans*." *Howard Hughes Research Fellowship Symposium*, Durham, NC. (Poster)
2015 "Detecting Antibiotic Resistance with dCas9." *iGEM regional conference*, College Park, MD. (Talk).

PROFESSIONAL ACTIVITIES & TEACHING EXPERIENCE

2025-2026 Primary research mentor for an undergraduate student in the Church laboratory (1 year)
2023-2024 Primary research mentor for an undergraduate student in the Zhang laboratory (1.5 years)
2023 Featured speaker, Biotech Engineering Exploration Challenge for High Schools
- Led a biotech-focused discussion for an initiative aimed at Boston-area high schoolers

2023 Presenter, Speed Science Workshop (McGovern Institute)
- Developed a 3D-printed interactive demo of my PhD work for MIT donors and administrators

2019-2020 Teaching assistant, *20.109 Laboratory Fundamentals in Biological Engineering*, MIT
- Taught undergraduates laboratory techniques in bioengineering

2016-2018 President, Duke American Society for Microbiology (ASM)
- Oversaw a student organization promoting interest in microbiology research